Anaphora Resolution on the GAP Coreference Dataset

Assignment 4 — Computational Linguistics / Natural Language Processing

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1. Task Formulation & Evaluation Regimes

The objective of this assignment is to resolve third-person singular personal and possessive pronouns (*he, his, him, she, her, hers*) to their correct antecedent in multi-sentence discourse segments extracted from Wikipedia articles. The dataset used is the Gender-Balanced Coreference (GAP) benchmark, containing 2,000 training, 454 validation, and 2,000 test segments. Each segment contains a snippet, target pronoun offset, two target named entities (Entity A and Entity B), and ground-truth coreference labels.

To ensure thorough evaluation, we report metrics under two standard experimental settings:

1. **Full POS Candidate Pool:** The model evaluates all candidate noun phrases extracted from the discourse prior to the pronoun. A prediction is scored as correct if its character span matches either gold entity.
2. **GAP Evaluation (Entity A vs. Entity B):** The model decides directly between the two candidate entities annotated in the benchmark, matching standard competitive benchmark scoring.

2. Problem-by-Problem Implementation & Results

Problem 1 (10 marks): Candidate Noun Phrase Extraction

Methodology: We segment the text into sentences using `nltk.sent_tokenize` and tokens via `nltk.word_tokenize`, followed by part-of-speech tagging using `nltk.pos_tag`. We formulate a Context-Free Grammar chunk parser with `nltk.RegexpParser`:

```
NP: {<DT>?<JJ.*>*<NN.*>+}      # Proper nouns, common noun phrases, modified
nouns
PP: {<IN><NP>}                   # Prepositional phrases
```

Only noun phrases whose end character offset strictly precedes the target pronoun are retained as valid candidate antecedents.

- Total test discourse segments processed: **2,000**
- Total candidate noun phrases extracted: **42,597**

- Average candidate NPs per segment: **21.30**

Problem 2 (5 marks): Random Antecedent Baseline (100 Runs)

Methodology: For every segment in the test set, an antecedent candidate is drawn uniformly at random. We run 100 independent trials and compute the average accuracy.

- Full POS Candidate Pool Accuracy: **7.13%** (consistently close to the theoretical 1 / 21.30 = 4.69% lower bound plus entity clustering)
- GAP Candidate Entities (A vs. B) Accuracy: **44.07%**

Problem 3 (5 marks): Most Recent Antecedent Baseline

Methodology: Selects the candidate noun phrase immediately preceding the target pronoun in linear character offset distance.

- Full POS Candidate Pool Accuracy: **12.85%** (257 / 2,000 correct)
- GAP Candidate Entities (A vs. B) Accuracy: **40.15%** (803 / 2,000 correct)

Observation: Pure linear recency is inadequate for full candidate resolution because the immediately preceding noun is frequently an inanimate object, prepositional complement, or modifying noun rather than the topical subject.

Problem 4 (10 marks): Custom Logistic Regression Classifier

Mathematical Formulation: Implemented from scratch using standard Python libraries and gradient descent:

$$P(y=1 \mid x) = \sigma(w^T x + b) = 1 / (1 + e^{-(w^T x + b)})$$

$$L(w, b) = - (1/N) \Sigma [w_{pos} y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + (\lambda/2) ||w||_2^2$$

Feature Engineering (12 Linguistic Features):

1. `entity_frequency`: Normalized occurrence count of candidate tokens in the preceding discourse.
2. `is_proper_noun`: Binary indicator (1 if tagged `NNP`/`NNPS`, 0 otherwise).
3. `grammatical_role_subj`: Binary indicator (1 if NP is the subject under an S node).
4. `recency_inv_rank`: Inverse distance rank (1 / rank).
5. `char_distance_closeness`: Linear distance normalized by text length (1 - dist / max_len).
6. `same_sentence`: 1 if the candidate and target pronoun share the same sentence.
7. `gender_match`: Heuristic match between candidate name/noun and pronoun gender.
8. `gender_mismatch`: Strict contradiction penalty (e.g., female name with 'he').
9. `number_agreement`: Grammatical plurality check.
10. `parallel_role`: 1 if both candidate and pronoun are syntactic subjects.
11. `is_hobbs_top1`: 1 if candidate is chosen first by Hobbs' tree walk.
12. `in_hobbs_top3`: 1 if candidate is within the top-3 Hobbs ranks.

Hyperparameter Grid Search (on 454 Development Segments):
We swept learning rate $\alpha \in \{0.01, 0.05, 0.1\}$, L2 penalty $\lambda \in \{0.0, 0.0001, 0.001\}$, and class weight $w_{pos} \in \{2.0, 4.0\}$.
Optimal Parameters: $\alpha = 0.1$, $\lambda = 0.001$, $w_{pos} = 4.0$ (achieved **62.39% F1** on GAP A/B validation).

Test Set Performance:

- Full Candidate Pool: Precision = **40.58%** | Recall = **45.57%** | **F1 = 42.93%**
- GAP Candidates (A vs. B): Precision = **56.65%** | Recall = **58.60%** | **F1 = 57.61%**

Problem 5 (5 marks): Learned Feature Weights & Top-5 Interpretation

Learned Intercept (Bias): **-6.6303** (accounts for the strong class imbalance where non-antecedents heavily outnumber antecedents).

Rank	Feature Name	Learned Weight	Linguistic Interpretation
1	**entity_frequency**	**+4.5122**	Topical Salience: Entities mentioned frequently throughout the paragraph represent central

Rank	Feature Name	Learned Weight	Linguistic Interpretation
			discourse topics, making them prime targets for pronominalization.
2	**is_proper_noun**	**+2.5635**	Named Entity Prior: Third-person singular pronouns in biographical narratives overwhelmingly refer to named persons rather than common nouns.
3	**grammatical_role_subj**	**+1.0529**	Syntactic Prominence: Syntactic subjects occupy the top position in Centering Theory's Forward-Looking Centers list (C_f).
4	**gender_mismatch**	**-0.9860**	Agreement Constraint: Substantial negative weight strongly penalizes gender contradictions, acting as a soft filter against impossible antecedents.
5	**is_hobbs_top1**	**+0.6022**	Structural Tree Validation: Confirms that configurational syntax traversal provides valuable structural signal complementary to statistical features.

Problem 6 (5 marks): Error Analysis & Improvements

We systematically analyzed the **960 test errors** produced by our classifier on gold-annotated segments. Four recurring error categories were identified:

1. **Kinship & Genitive Relational Constructions (e.g., test-1):**
   Context: "...Bob Suter are Dehner's uncles. His cousin is Minnesota Wild's..."
   Prediction: 'Former NHLer Gary Suter' (True: 'Dehner')
   Cause: The classifier overweights recent subject proper nouns and misses the genitive head modifier relationship (`Dehner's uncles`).
   Proposed Improvement: Incorporate typed dependency parse relations (`nmod:poss`) and semantic kinship roles.
2. **Direct Speech & Quotation Attribution Frames (e.g., test-4):**
   Context: "At the trial, Pisciotta said: '... Marchesano and also Signor Scelba ...' During his trial..."
   Prediction: 'Marchesano' (True: 'Pisciotta')
   Cause: Mentions inside long quoted dialogue artificially inflated candidate frequency and recency.
   Proposed Improvement: Segment direct speech boundaries and apply a speaker-continuation heuristic for narrative framing.
3. **Parenthetical Appositives & Aliases (e.g., test-5):**
   Context: "Eddie Devane (William McNamara) is a young sailor... during his service."
   Prediction: 'William McNamara' (True: 'Eddie')
   Cause: Shallow POS chunking cannot differentiate the real syntactic subject from parenthetical actor names.
   Proposed Improvement: Detect parentheses explicitly and assign lower salience weights to parenthetical alias strings.
4. **Discourse Cohesion & Event Semantics (e.g., test-8):**
   Context: "...bites Jeni's tongue... Sophie... pulls her in."
   Prediction: 'Sophie' (True: 'Jeni')
   Cause: Both candidates share feminine gender; purely syntactic cues cannot resolve the agent-patient relation.
   Proposed Improvement: Integrate selectional restrictions on verbs and entity-grid discourse cohesion models.

Problem 7 (10 marks): Rule-Based Hobbs' Algorithm

Methodology: We implemented the classical 9-step configurational search algorithm (Hobbs, 1978) on NLTK `ParentedTree` parse representations. The algorithm begins at the pronoun's immediate NP parent, traverses upwards to the first governing NP or S node, searches subtrees left-to-right below the node, and systematically sweeps preceding sentence trees breadth-first while enforcing gender and number agreement.

- Full Candidate Pool: Precision = **30.42%** | Recall = **34.29%** | **F1 = 32.24%**

- GAP Candidates (A vs. B): Precision = **43.46%** | Recall = **42.36%** | **F1 = 42.90%**

Problem 8 (5 marks): Comparison & Synthesis

Model / Method	Full-Pool F1	GAP (A/B) Prec	GAP (A/B) Rec	GAP (A/B) F1
Random Assignment Baseline (100 runs)	7.13%	44.07%	44.07%	44.07%
Most Recent Antecedent Baseline	12.85%	40.15%	40.15%	40.15%
Rule-Based Hobbs' Algorithm	32.24%	43.46%	42.36%	42.90%
Custom Logistic Regression (Ours)	**42.93%**	**56.65%**	**58.60%**	**57.61%**
Extra Credit: Scikit-Learn LR	*42.33%*	*55.61%*	*59.22%*	*57.36%*

Qualitative Insights:

- *Hobbs vs. Baselines:* Hobbs substantially improves over random and recency baselines (+19.39% on Full Pool F1) because syntactic dominance (commanding S/NP nodes) is far more reliable than linear word proximity.
- *Logistic Regression vs. Hobbs:* Our custom classifier outperforms Hobbs by **+14.71% F1 on GAP A/B** and **+10.69% F1 on the full pool**. While Hobbs follows a fixed deterministic sequence, Logistic Regression flexibly trades off entity salience, naming status, distance, and syntactic configuration simultaneously.

Extra Credit (10 marks): Scikit-Learn Benchmark Validation

We implemented a reference classifier using `sklearn.linear_model.LogisticRegression` with an L2 penalty and balanced class weighting. Grid search on the development set yielded optimal parameters of C = 0.1 and pos_weight = 4.0. On the test set, scikit-learn achieved **57.36% GAP F1** and **42.33% Full-Pool F1**, closely matching our from-scratch model (**57.61%** and **42.93%**), rigorously verifying the numerical stability and mathematical correctness of our custom implementation.